

Tutorial 1 - Answers

Question 1

Inductive Statement $S(n)$: “If there are n people on Earth, all of them hate cookies.”

Base case $S(0)$: This case is vacuously true, since there is no one on Earth to violate it.

Inductive hypothesis: If there are n people on Earth, they all hate cookies.

What went wrong? The proof’s trick needs two different people to work, one goes to the Moon, another goes to Mars. This only works if there are at least 2 people to begin with! The only proven starting point is 0 people (which is useless). If we try to jump from 0 people to 1 person, there is no second person available to send to Mars, so the trick cannot be even used. Since the very first jump breaks, the chain never really gets going. It could work fine for more people but it never gets that chance to do so. The only thing proven is the empty useless case of 0 people.

I love cookies (those from Costco are goated)

Question 2

Part 1: WOP from POSI

Proof. Suppose for contradiction that there exists a subset $A \subseteq \mathbb{N}$ such that A is nonempty and has no least element. Define $S(n) : n \notin A$ for all $n \in \mathbb{N}$. Prove by strong induction that $S(n)$ holds for all $n \in \mathbb{N}$.

Base Case ($n = 0$). Suppose for contradiction that $0 \in A$. Since $0 \leq x$ for every natural number x , 0 would be the least element of A . This contradicts the assumption that A has no least element. Therefore $0 \notin A$.

Inductive step. Let $n \in \mathbb{N}$ and assume as the strong induction hypothesis that $k \notin A$ for every natural number k such that $k < n$. We will show that $n \notin A$. Suppose for contradiction that $n \in A$.

By the inductive hypothesis, no natural number strictly smaller than n belongs to A . Consequently, for any element $a \in A$, we cannot have $a < n$, which implies that $a \geq n$ for all $a \in A$. Because we assume $n \in A$ and every other element in A is greater than or equal to n , $n = \min A$. This contradicts our assumption.

Therefore, the supposition that $n \in A$ must be false, meaning $n \notin A$.

By POSI, $n \notin A$ for all $n \in \mathbb{N}$. This implies that A contains no elements. However, this contradicts our initial premise that A is nonempty. Thus, every nonempty subset of $\mathbb{N}$ must have a least element. ■

Part 2: POMI from WOP

Theorem. (Principle of Mathematical Induction) Let $P(n)$ be a statement defined for each $n \in \mathbb{N}$. Suppose: 1. $P(0)$ is true, and 2. for every $k \geq 0$, $P(k) \implies P(k+1)$

Then $P(n)$ is true for all $n \in \mathbb{N}$.

Proof. We use the Well-Ordering Principle. Assume that (1) and (2) hold. Suppose, for contradiction, that $P(n)$ fails for some n , and let

$$S = \{n \in \mathbb{N} : P(n) \text{ is false}\}.$$

By assumption $S \neq \emptyset$, so by WOP, S has a least element m .

- $m \neq 0$. By (1), $P(0)$ is true, so $0 \notin S$. Hence $m \geq 1$, so $m-1 \in \mathbb{N}$.
- $P(m-1)$ is true. Since $m-1 < m$ and m is the least element of S , we have $m-1 \notin S$, i.e. $P(m-1)$ is true.
- $P(m)$ is true. By (2) with $k = m-1$: $P(m-1) \implies P(m)$. So $P(m)$ is true, which contradicts $m \in S$.

Hence $S = \emptyset$, that is, $P(n)$ is true for all $n \in \mathbb{N}$. ■

Question 3

Proof. Suppose for contradiction that the claim is false. Let

$$T = \{n \in \mathbb{N} : n \geq 2 \text{ and } n \text{ is not a product of primes}\}$$

By assumption, T is nonempty, so by WOP, let $n_0 = \min T$. Since n_0 is the minimal counterexample, every integer k with $2 \leq k < n_0$ can be written as a product of primes.

Case 1: n_0 is prime. Then n_0 is trivially a product of primes (itself), contradiction $n_0 \in T$.

Case 2: n_0 is not prime. Then $n_0 = ab$ for some integers a, b with $2 \leq a < n_0$ and $2 \leq b < n_0$. Since $a, b < n_0$, minimality of n_0 gives that a and b are each products of primes, say

$$a = p_1 p_2 \dots p_k, \quad b = q_1 q_2 \dots q_\ell.$$

Then

$$n_0 = ab = (p_1 p_2 \dots p_k)(q_1 q_2 \dots q_\ell)$$

so n_0 is a product of primes, contradicting $n_0 \in T$.

Both cases contradict $n_0 \in T$. Hence $T = \emptyset$, so every integer $n \geq 2$ can be written as a product of primes. ■

Question 4

Exercise 2.29

Proof. We will proceed with induction on n .

Base case ($n = 1$). A 2×2 grid with one square removed has exactly 3 remaining squares, which form a single triomino.

Inductive step. Assume the claim holds for n . Consider a $2^{n+1} \times 2^{n+1}$ grid with one squared removed. Partition it into four $2^n \times 2^n$ quadrants. The removed square lies in exactly one quadrant; call it Q_1 , and label the other three Q_2, Q_3, Q_4 .

Place a single triomino at the center of the grid covering exactly one cell from each of Q_2, Q_3, Q_4 , the three cells adjacent to the grid's centre point, in the three quadrants not containing the removed square. By rotational symmetry, this placement is possible no matter which quadrant Q_1 is.

After this placement, all four quadrants Q_1 (missing its original square) and Q_2, Q_3, Q_4 (each missing the cell just covered) are $2^n \times 2^n$ grids with one square removed. By the inductive hypothesis, each can be tiled by triominoes. Together with the central triomino, this tiles the full $2^{n+1} \times 2^{n+1}$ grid. ■

2^n cube extension

Claim. For every $n \geq 1$, a cube of side 2^n can be completely tiled (no gaps, no overlaps) by tetraminos, where a tetramino is an L-triomino with one extra cube attached above its middle cube.

(This requires $n \geq 1$: a cube of side 2^n has 2^{3n} unit cubes, and we need $4 \mid 2^{3n}$, which holds for exactly when $n \geq 1$.)

Proof. We will proceed by induction on n .

Base case ($n = 1$). Tile the $2 \times 2 \times 2$ cube (8 unit cubes) with 2 tetraminos. Label cells by $(x, y, z) \in \{0, 1\}^3$.

- Piece A: L-triomino $\{(0, 0, 0), (1, 0, 0), (0, 1, 0)\}$, middle cell $(0, 0, 0)$ plus extra cube $(0, 0, 1)$ directly above the middle.
- Piece B: L-triomino $\{(1, 0, 1), (0, 1, 1), (1, 1, 1)\}$, middle cell $(1, 1, 1)$ plus extra cube $(1, 1, 0)$ directly below the middle.

Together A and B cover all 8 cells exactly once. Rotating Piece A by 180° about the axis through the cube's centre in direction $(1, -1, 0)$ maps it exactly onto Piece B, confirming both are genuine rotations of the same tetramino (no reflection needed).

Inductive step. Assume a cube of side 2^n can be tiled by tetraminos. A cube of side 2^{n+1} splits exactly into eight sub-cubes of side 2^n (a $2 \times 2 \times 2$ block of octants), with no leftover cells. By the inductive hypothesis, each sub-cube can be tiled by tetraminos; their union tiles the full 2^{n+1} cube.

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Question 5

We define $S(n)$ such that

$$S(n) : x_n = \frac{\varphi^n - (1 - \varphi)^n}{\sqrt{5}}$$

Start by letting $\varphi = \frac{a}{b}$. Therefore

$$\varphi = 1 + \frac{1}{\varphi} \implies \varphi^2 = \varphi + 1 \implies \varphi^2 - \varphi - 1 = 0 \quad (1)$$

If we solve (1) we will get two roots, φ and $1 - \varphi$. Let r be either root. Since each root satisfies the quadratic, it will satisfy this equation as well:

$$r^2 = r + 1$$

There is a bit of a problem here as the identity only talks about the square of r . We want something general, that is, an expression for r^n for some arbitrary n . We will take $r^2 = r + 1$ and multiply both sides by r^{k-1} for any $k \geq 1$:

$$r^2 \cdot r^{k-1} = (r + 1) \cdot r^{k-1}$$

By exponent rules we have:

$$r^{k+1} = r^k + r^{k-1} \quad (2)$$

This is what we wanted as it connects three consecutive powers for any k . It holds for $r = \varphi$ and $r = 1 - \varphi$.

We will now use strong induction on $S(n)$ with two bases cases since (2) looks back two steps.

Base Cases. Let $n = 0$:

$$x_0 = \frac{\varphi^0 - (1 - \varphi)^0}{\sqrt{5}} = \frac{1 - 1}{\sqrt{5}} = 0 \in \mathbb{N} \implies S(0) \text{ holds.}$$

Let $n = 1$:

$$x_1 = \frac{\varphi^1 - (1 - \varphi)^1}{\sqrt{5}} = \frac{2\varphi - 1}{\sqrt{5}} = \frac{\sqrt{5}}{\sqrt{5}} = 1 \in \mathbb{N} \implies S(1) \text{ holds.}$$

Inductive hypothesis. Assume $S(j)$ holds for every $j \leq k$, that is $x_0, x_1, \dots, x_k \in \mathbb{N}$.

Inductive step. We show that $S(k + 1)$ holds.

$$x_{k+1} = \frac{\varphi^{k+1} - (1 - \varphi)^{k+1}}{\sqrt{5}}$$

Applying (2) to each term:

$$\begin{aligned} &= \frac{(\varphi^k + \varphi^{k-1}) - ((1 - \varphi)^k + (1 - \varphi)^{k-1})}{\sqrt{5}} \\ &= \underbrace{\frac{\varphi^k - (1 - \varphi)^k}{\sqrt{5}}}_{x_k} + \underbrace{\frac{\varphi^{k-1} - (1 - \varphi)^{k-1}}{\sqrt{5}}}_{x_{k-1}} = x_k + x_{k-1} \end{aligned}$$

By the inductive hypothesis, both x_k and x_{k-1} are natural numbers. Since $\mathbb{N}$ is closed under addition, $x_{k+1} \in \mathbb{N}$. Thus S_{k+1} holds.

By the principle of strong induction $S(n)$ holds for all $n \in \mathbb{N}$.

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